

Reading Notes: Government Bond Risk and Return in the US and China

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My study notes (models + table/figure interpretation)

Contents

1	Executive summary	3
1.1	The organizing question	3
1.2	Core idea in one equation	3
1.3	Main empirical claims I take away	3
2	Data and factor construction (what is being modeled)	4
2.1	Implied zero-coupon returns	4
2.2	Standardization and PCA	4
2.3	Interpretation of the first two factors	4
3	Theory: continuous-time asset pricing backbone (Eqs. (1)–(13))	5
3.1	Real asset price dynamics	5
3.2	Portfolio returns and the no-arbitrage restriction	5
3.3	Stochastic discount factor representation	5
3.4	Link to preference-based models	6
3.5	Nominal pricing: inflation as locally riskless	6
3.6	Two priced bond factors	6
4	Empirical model and estimation (Eqs. (14)–(28))	7
4.1	Discrete-time factor return model	7
4.2	State dependence via linear specifications	7
4.3	Estimation equations	8

4.4	Moment conditions and GMM	8
4.5	ZLB-augmented specification	8
4.6	Goodness-of-fit measures	9
5	Risk-premium decomposition and mapping to bond maturities (Eqs. (29)–(33))	9
5.1	Variance decomposition of risk premia	9
5.2	From factor dynamics to maturity-specific bond dynamics	9
6	Robustness: a GARCH-in-mean alternative (Eqs. (34)–(35))	10
7	Interpreting the tables (Table 1–Table 6)	10
7.1	Table 1: Factor structure and performance of implied zeros	10
7.2	Table 2: Factor structure and performance of UST ETFs	12
7.3	Table 3: GMM estimates of US factor dynamics (baseline, no ZLB shift)	12
7.4	Table 4: US factor dynamics allowing ZLB effects	14
7.5	Table 5: Monthly versus annual R^2 in bond return regressions	16
7.6	Table 6: GMM estimates of China factor dynamics	16
8	Interpreting the figures (Fig. 1–Fig. 8)	18
8.1	Fig. 1: China bond market capitalization	18
8.2	Fig. 2: MOVE index (US interest rate volatility)	18
8.3	Fig. 3: US Factor 1 dynamics (volatility, Sharpe ratio, risk premium)	19
8.4	Fig. 4: US Factor 2 dynamics	19
8.5	Fig. 5: US bond dynamics (2-year and 10-year implied zeros, baseline)	21
8.6	Fig. 6: US bond dynamics with ZLB effects	21
8.7	Fig. 7: China Factor 1 and Factor 2 dynamics	22
8.8	Fig. 8: China bond dynamics (2-year and 10-year implied zeros)	22
9	Concluding synthesis (my coherent mental model)	23
9.1	What I would say if asked “what is the paper really doing?”	23

1 Executive summary

1.1 The organizing question

My organizing question while reading is:

How can I characterize government bond “risk and return” using a decomposition that separates (i) *how much risk there is* (quantity of risk / volatility) from (ii) *how the market prices that risk* (price of risk / Sharpe ratio), and how do these objects differ across the US and China?

1.2 Core idea in one equation

For a factor portfolio j , the paper targets the identity

$$\text{risk premium}_{j,t} \approx \sigma_{j,t} \theta_{j,t},$$

i.e., *expected excess return equals volatility times Sharpe ratio*. The central empirical message is that both $\sigma_{j,t}$ and $\theta_{j,t}$ vary over time, and their co-movement differs sharply between US Treasuries and Chinese government bonds.

1.3 Main empirical claims I take away

- **Bond returns are well summarized by a small number of priced factors.** The first two PCA factors (level-like and slope-like) account for the dominant variation in standardized bond excess returns.
- **US: volatility and Sharpe ratio are often positively correlated.** This makes bond risk premia move strongly with volatility in many periods, but this relationship becomes different around the ZLB¹.
- **China: volatility and Sharpe ratio are often negatively correlated.** This is a key cross-country difference: increases in risk (volatility) need not coincide with increases in compensation per unit risk (Sharpe).
- **Methodological warning: overlapping annual return regressions can inflate R^2 .** The paper argues that many “high R^2 ” bond predictability results are mechanical when overlapping annual returns are used.

¹The Zero Lower Bound (ZLB) is a macroeconomic situation where a central bank’s nominal interest rate is at or near zero percent. As a constraint on conventional monetary policy, the ZLB limits the ability to stimulate economic growth via rate cuts, often resulting in a “liquidity trap” where further cuts are ineffective, as people prefer holding cash.

2 Data and factor construction (what is being modeled)

2.1 Implied zero-coupon returns

The empirical objects of interest are monthly *excess returns* on implied zero-coupon bonds:

- US Treasury implied zero-coupon yields are constructed from a yield curve fitted to key maturities (par yields), then converted into monthly implied zero-coupon bond prices and returns.
- China government bond (CGB) implied zero-coupon yields are constructed analogously from Chinese yield curve inputs.
- The paper focuses on annual maturities from 1 to 10 years.

2.2 Standardization and PCA

A technical but important step: before PCA, the paper standardizes each maturity’s excess return series to have unit volatility, and then applies PCA to the resulting correlation matrix. My intuition:

- Without standardization, long-maturity returns (higher volatility) would dominate PCA mechanically.
- With standardization, PCA finds *co-movement patterns* rather than simply “which maturity is most volatile.”

Let $z_{i,t}$ be the monthly excess return of maturity i (annualized or not depending on the paper’s convention). Let v_i be the sample volatility of $z_{i,t}$. The standardized return is

$$sz_{i,t} \equiv \frac{z_{i,t}}{v_i}.$$

PCA is applied to the vector $sz_t = (sz_{1,t}, \dots, sz_{10,t})^\top$; factor portfolios correspond to eigenvectors of the correlation matrix.

2.3 Interpretation of the first two factors

I interpret the first two PCA factors as:

- **Factor 1 (dominant):** a *level-like* factor (similar sign across maturities).
- **Factor 2:** a *slope-like / duration spread* factor (long-short across maturities).

The paper uses these two priced factors to build a *two-factor risk-return decomposition*.

3 Theory: continuous-time asset pricing backbone (Eqs. (1)–(13))

3.1 Real asset price dynamics

The starting point is a generic diffusion model for *real* asset prices $S_{i,t}$:

$$\frac{dS_{i,t}}{S_{i,t}} = \mu_{i,t} dt + \sigma_{i,t} dB_t, \quad i = 1, \dots, n, \quad (1)$$

where B_t is an m -dimensional Brownian motion, and $\sigma_{i,t}$ is the (row) vector of loadings on Brownian shocks.

3.2 Portfolio returns and the no-arbitrage restriction

For a self-financing portfolio with wealth W_t and holdings π_t in risky assets:

$$dW_t = \left(r_t W_t + \pi_t (\mu_t - r_t \mathbf{1}) \right) dt + \pi_t \sigma_t dB_t. \quad (2)$$

No-arbitrage implies the existence of a market price of risk vector θ_t satisfying:

$$\sigma_t \theta_t = \mu_t - r_t \mathbf{1}. \quad (3)$$

Proposition 1 (Excess return decomposition). *Under (1)–(3), each asset's excess return can be written as*

$$\frac{dS_{i,t}}{S_{i,t}} - r_t dt = \sigma_{i,t} \theta_t dt + \sigma_{i,t} dB_t. \quad (4)$$

Proof. Start from (1): $dS_{i,t}/S_{i,t} = \mu_{i,t} dt + \sigma_{i,t} dB_t$. Subtract $r_t dt$ to obtain $dS_{i,t}/S_{i,t} - r_t dt = (\mu_{i,t} - r_t) dt + \sigma_{i,t} dB_t$. By (3), the scalar drift $(\mu_{i,t} - r_t)$ equals $\sigma_{i,t} \theta_t$ (the i th row of $\sigma_t \theta_t$), so the drift becomes $\sigma_{i,t} \theta_t dt$, which yields (4). $\square$

3.3 Stochastic discount factor representation

The paper writes the (real) stochastic discount factor (SDF) as:

$$M_t = \exp \left(- \int_0^t r_s ds - \int_0^t \theta_s^\top dB_s - \frac{1}{2} \int_0^t \theta_s^\top \theta_s ds \right). \quad (5)$$

Then the generic pricing relation is:

$$S_{i,t} = \mathbb{E}_t \left(\frac{M_u}{M_t} S_{i,u} \right), \quad u > t. \quad (6)$$

3.4 Link to preference-based models

In equilibrium, the market price of risk can be written as:

$$\theta_t = R_t \sigma_{c,t}, \quad (7)$$

where $\sigma_{c,t}$ is consumption growth volatility and R_t is risk aversion (possibly time-varying). My interpretation: (7) explains why θ_t can be state-dependent even if fundamentals are smooth.

3.5 Nominal pricing: inflation as locally riskless

Let q_t be the price level (inflation index) that follows:

$$\frac{dq_t}{q_t} = i_t dt, \quad (8)$$

with no diffusion term (“locally riskless inflation”).

Nominal prices satisfy $P_{i,t} = q_t S_{i,t}$. Then the paper shows:

$$\frac{dP_{i,t}}{P_{i,t}} - (r_t + i_t)dt = \frac{dS_{i,t}}{S_{i,t}} - r_t dt = \sigma_{i,t} \theta_t dt + \sigma_{i,t} dB_t. \quad (9)$$

Proposition 2 (Why (9) holds). *If q_t has no diffusion term as in (8), then the nominal excess return of $P_{i,t} = q_t S_{i,t}$ equals the real excess return of $S_{i,t}$.*

Proof. Using Itô: $dP/P = d(qS)/(qS) = dq/q + dS/S + (dq/q)(dS/S)$. Because dq/q has only a dt term, the quadratic covariation term is zero. Thus $dP/P = i_t dt + \mu_{i,t} dt + \sigma_{i,t} dB_t$. Subtract the nominal short rate $(r_t + i_t)dt$ and use $\mu_{i,t} - r_t = \sigma_{i,t} \theta_t$ from (3), yielding (9). $\square$

The nominal pricing kernel is therefore:

$$\frac{M_t}{q_t} = \exp\left(-\int_0^t (r_s + i_s) ds - \int_0^t \theta_s^\top dB_s - \frac{1}{2} \int_0^t \theta_s^\top \theta_s ds\right), \quad (10)$$

and a nominal zero-coupon bond price is:

$$P_t^T = \mathbb{E}_t\left\{\exp\left(-\int_t^T (r_s + i_s) ds - \int_t^T \theta_s^\top dB_s - \frac{1}{2} \int_t^T \theta_s^\top \theta_s ds\right)\right\}. \quad (11)$$

3.6 Two priced bond factors

The paper summarizes government bond excess returns using two factor portfolios $F_{j,t}$:

$$dF_{j,t} = \sigma_{j,t} \theta_{j,t} dt + \sigma_{j,t} dB_{j,t}, \quad j = 1, 2, \quad (12)$$

where:

- $\sigma_{j,t}$ is the **quantity of risk** for factor j ,
- $\theta_{j,t}$ is the **price of risk** (factor Sharpe ratio),
- $\sigma_{j,t}\theta_{j,t}$ is the **factor risk premium**.

Finally, the (implied) zero-coupon bond excess returns are spanned by the factors:

$$\frac{dP_{i,t}}{P_{i,t}} - (r_t + i_t)dt = \beta_{i,1}dF_{1,t} + \beta_{i,2}dF_{2,t}, \quad i = 1, \dots, 10. \quad (13)$$

My mental model: $\beta_{i,1}$ and $\beta_{i,2}$ are essentially the PCA loadings (appropriately normalized) mapping factor risk premia into maturity-specific risk premia.

4 Empirical model and estimation (Eqs. (14)–(28))

4.1 Discrete-time factor return model

The monthly discrete-time analogue of (12) is:

$$R_{j,t+1} = \sigma_{j,t}\theta_{j,t} + \sigma_{j,t}\varepsilon_{j,t+1}, \quad \mathbb{E}_t[\varepsilon_{j,t+1}] = 0, \quad \text{Var}_t(\varepsilon_{j,t+1}) = 1. \quad (14)$$

The object of interest is the conditional mean:

$$\mathbb{E}_t[R_{j,t+1}] = \sigma_{j,t}\theta_{j,t},$$

so expected returns are *bilinear* in $\sigma_{j,t}$ and $\theta_{j,t}$.

4.2 State dependence via linear specifications

They assume both the factor volatility and factor Sharpe ratio are linear in predictors X_t :

$$\sigma_{j,t} = X_t\beta_j^\sigma, \quad (15)$$

$$\theta_{j,t} = X_t\beta_j^\theta. \quad (16)$$

Important implication: $\mathbb{E}_t[R_{j,t+1}]$ is quadratic in X_t :

$$\mathbb{E}_t[R_{j,t+1}] = (X_t\beta_j^\sigma)(X_t\beta_j^\theta).$$

4.3 Estimation equations

They write:

$$R_{j,t+1} = \alpha_j + (X_t \beta_j^\sigma)(X_t \beta_j^\theta) + u_{j,t+1}, \quad (17)$$

with $u_{j,t+1} \equiv \sigma_{j,t} \varepsilon_{j,t+1}$.

To estimate $\sigma_{j,t}$ from returns, they use a scaled absolute residual:

$$\sqrt{\frac{\pi}{2}} |u_{j,t+1}| = X_t \beta_j^\sigma + v_{j,t+1}. \quad (18)$$

Remark 1 (Why the $\sqrt{\pi/2}$ scaling?). If $\varepsilon \sim \mathcal{N}(0, 1)$, then $|\varepsilon|$ has mean $\sqrt{2/\pi}$. Thus $\sqrt{\pi/2} |\varepsilon|$ has mean 1. Since $u = \sigma \varepsilon$, we have $\sqrt{\pi/2} |u|$ centered at σ . Equation (18) uses this property to form an approximately unbiased volatility proxy.

4.4 Moment conditions and GMM

Let Z_t collect instruments that include the unique elements of $X_t^\top X_t$ (so that quadratic terms are “in the instrument space”). The GMM moment conditions are:

$$\mathbb{E}[u_{j,t+1} Z_t] = 0, \quad (19)$$

$$\mathbb{E}[v_{j,t+1} X_t] = 0. \quad (20)$$

4.5 ZLB-augmented specification

To allow a regime shift at/near the zero lower bound, define an indicator L_t (the paper uses an effective fed funds rule-based definition). The augmented return equation is:

$$R_{j,t+1} = \alpha_j + \delta_j L_t + \left(X_t \beta_j^\sigma + L_t X_t \gamma_j^\sigma \right) \left(X_t \beta_j^\theta + L_t X_t \gamma_j^\theta \right) + u_{j,t+1}, \quad (21)$$

and the volatility equation becomes:

$$\sqrt{\frac{\pi}{2}} |u_{j,t+1}| = X_t \beta_j^\sigma + L_t X_t \gamma_j^\sigma + v_{j,t+1}. \quad (22)$$

Here γ_j^σ and γ_j^θ capture *differences in the volatility and Sharpe-ratio laws of motion* during ZLB periods.

The corresponding moment conditions are:

$$\mathbb{E}[u_{j,t+1}Z_t] = 0, \quad (23)$$

$$\mathbb{E}[v_{j,t+1}X_t] = 0, \quad (24)$$

$$\mathbb{E}[u_{j,t+1}Z_tL_t] = 0, \quad (25)$$

$$\mathbb{E}[v_{j,t+1}X_tL_t] = 0. \quad (26)$$

4.6 Goodness-of-fit measures

They report two pseudo- R^2 measures:

$$\text{GoF}(1) = 1 - \frac{\sum_t v_{j,t+1}^2}{\frac{\pi}{2} \sum_t (|u_{j,t+1}| - \overline{|u_j|})^2}, \quad (27)$$

$$\text{GoF}(2) = 1 - \frac{\sum_t u_{j,t+1}^2}{\sum_t (R_{j,t+1} - \bar{R}_j)^2}. \quad (28)$$

My interpretation: GoF(1) is about explaining volatility dynamics; GoF(2) is about explaining return dynamics.

5 Risk-premium decomposition and mapping to bond maturities (Eqs. (29)–(33))

5.1 Variance decomposition of risk premia

Let the factor risk premium be $\sigma_t\theta_t$. A first-order approximation yields:

$$(\sigma_t\theta_t - \overline{\sigma\theta})^2 \approx \bar{\theta}^2(\sigma_t - \bar{\sigma})^2 + \bar{\sigma}^2(\theta_t - \bar{\theta})^2 + 2\bar{\sigma}\bar{\theta}(\sigma_t - \bar{\sigma})(\theta_t - \bar{\theta}). \quad (29)$$

I read (29) as:

- the first term: contribution of volatility variation holding average Sharpe ratio fixed,
- the second term: contribution of Sharpe-ratio variation holding average volatility fixed,
- the third term: contribution of covariance (co-movement) between volatility and Sharpe ratio.

5.2 From factor dynamics to maturity-specific bond dynamics

Let $sz_{i,t}$ denote the standardized excess return on maturity i :

$$sz_{i,t} = \beta_{i,1}R_{1,t} + \beta_{i,2}R_{2,t}. \quad (30)$$

Since $z_{i,t} = v_i s z_{i,t}$, maturity- i volatility (annualized) is:

$$\text{Vol}_t(z_i) = \sqrt{12} v_i \sqrt{\beta_{i,1}^2 \sigma_{1,t}^2 + \beta_{i,2}^2 \sigma_{2,t}^2}. \quad (31)$$

The (annualized) risk premium is:

$$rp_t(z_i) = 12 v_i^2 \left(\beta_{i,1}^2 \sigma_{1,t}^2 \theta_{1,t} + \beta_{i,2}^2 \sigma_{2,t}^2 \theta_{2,t} \right), \quad (32)$$

and the corresponding Sharpe ratio is:

$$SR_t(z_i) = \sqrt{12} v_i \frac{\beta_{i,1}^2 \sigma_{1,t}^2 \theta_{1,t} + \beta_{i,2}^2 \sigma_{2,t}^2 \theta_{2,t}}{\sqrt{\beta_{i,1}^2 \sigma_{1,t}^2 + \beta_{i,2}^2 \sigma_{2,t}^2}}. \quad (33)$$

Remark 2 (Reading (31)–(33)). *Once I have estimates of $(\sigma_{1,t}, \theta_{1,t}, \sigma_{2,t}, \theta_{2,t})$ and the loadings $\beta_{i,1}, \beta_{i,2}$, I can produce time series for each maturity's:*

volatility vs. Sharpe ratio vs. risk premium.

This is exactly what the paper plots for 2-year and 10-year bonds in the US and China.

6 Robustness: a GARCH-in-mean alternative (Eqs. (34)–(35))

For robustness, they also estimate a GARCH(1,1)-M style volatility model:

$$R_{j,t} = (X_t \beta_j^\theta) \sigma_{j,t-1} + \varepsilon_{j,t}, \quad \varepsilon_{j,t} \sim \mathcal{N}(0, \sigma_{j,t-1}), \quad (34)$$

with conditional variance recursion:

$$\sigma_{j,t}^2 = e^{X_t \beta_j^\sigma} + \rho \sigma_{j,t-1}^2 + \lambda \varepsilon_{j,t}^2. \quad (35)$$

My interpretation: this keeps the same economic idea (expected return depends on volatility and Sharpe ratio), but allows volatility persistence via a standard latent-variance recursion.

7 Interpreting the tables (Table 1–Table 6)

7.1 Table 1: Factor structure and performance of implied zeros

What the table contains (how I read it).

- **Panel A:** factor portfolio weights (PCA eigenvectors) and percentage of total variance explained by Factors 1–3, for UST and CGB samples.

- **Panel B:** mean, volatility, and Sharpe ratio of each maturity’s implied zero excess return, plus the same moments for factor portfolios, across subsamples.

Table 1: Factor structure and performance of UST and CGB implied zero excess returns.

	UST Implied Zeros						CGB Implied Zeros		
	7/1976–12/1989			1/1990–12/2022			5/2004–12/2022		
	F1	F2	F3	F1	F2	F3	F1	F2	F3
A. Factor Structure									
Factor Var. as % of Tot.	94.69	4.07	0.73	90.85	7.18	1.42	82.36	13.54	2.46
Factor Vol	10.66	2.21	0.93	10.44	2.93	1.31	9.94	4.03	1.72
Factor SR	0.27	0.45	0.45	0.64	0.73	0.51	0.51	0.14	0.02
1-year zero loadings	0.30	0.57	0.43	0.26	0.66	0.65	0.25	0.54	−0.60
2-year zero loadings	0.31	0.40	0.11	0.31	0.41	−0.24	0.29	0.45	−0.11
3-year zero loadings	0.32	0.27	−0.14	0.32	0.24	−0.37	0.32	0.30	0.25
4-year zero loadings	0.32	0.15	−0.29	0.33	0.09	−0.33	0.34	0.16	0.38
5-year zero loadings	0.32	0.02	−0.39	0.33	−0.03	−0.23	0.34	0.02	0.39
6-year zero loadings	0.32	−0.14	−0.33	0.33	−0.12	−0.11	0.34	−0.11	0.23
7-year zero loadings	0.32	−0.26	−0.19	0.33	−0.20	0.01	0.33	−0.22	0.00
8-year zero loadings	0.32	−0.31	0.02	0.32	−0.26	0.13	0.33	−0.28	−0.14
9-year zero loadings	0.32	−0.34	0.27	0.32	−0.31	0.25	0.32	−0.33	−0.26
10-year zero loadings	0.31	−0.34	0.58	0.31	−0.34	0.36	0.30	−0.36	−0.36
B. Performance Measures									
	Mean	Vol	SR	Mean	Vol	SR	Mean	Vol	SR
1-year zero	1.40	2.51	0.56	0.69	0.66	1.05	0.37	0.88	0.42
2-year zero	1.56	4.70	0.33	1.30	1.66	0.78	0.81	1.51	0.54
3-year zero	1.68	6.34	0.26	1.76	2.70	0.65	1.08	2.09	0.52
4-year zero	1.94	8.07	0.24	2.33	3.71	0.63	1.33	2.69	0.49
5-year zero	2.26	9.71	0.23	2.76	4.70	0.59	1.61	3.35	0.48
6-year zero	2.61	11.11	0.23	3.24	5.62	0.58	2.02	3.87	0.52
7-year zero	2.60	12.43	0.21	3.38	6.51	0.52	1.95	4.45	0.44
8-year zero	2.73	13.55	0.20	3.63	7.37	0.49	2.11	5.03	0.42
9-year zero	2.83	14.46	0.20	3.79	8.24	0.46	2.30	5.64	0.41
10-year zero	2.84	15.25	0.19	3.77	9.11	0.41	2.43	6.29	0.39

Key interpretation.

- **A small number of factors dominates.** Factor 1 explains the vast majority of standardized return variation; Factor 2 still matters, especially in more recent samples and in China.
- **Sharpe ratios decline with maturity (“betting-against-duration” style pattern).** Shorter maturities have higher Sharpe ratios than longer maturities in the US.
- **Why the decline happens (mechanism via Factor 2).** Factor 2 has a relatively high Sharpe ratio, and short maturities load positively on it while long maturities load negatively. As maturity increases, the net exposure to the high-Sharpe Factor 2 decreases, pushing down the maturity Sharpe ratio.
- **Cross-country difference.** In China, the factor structure and the Sharpe ratios of factors

look different (Factor 2 appears less “priced” in the same way), motivating a separate dynamics analysis for CGB factors.

7.2 Table 2: Factor structure and performance of UST ETFs

Table 2: Factor structure and performance of UST ETF excess returns.

	F1	F2	F3
A. Factor Structure			
Factor Var. as % of Tot.	78.53	14.79	5.17
Factor Vol	7.52	3.26	1.93
Factor SR	0.58	0.79	0.44
0–1-year ETF	0.30	0.74	0.59
1–3-year ETF	0.40	0.39	−0.51
3–7-year ETF	0.44	0.05	−0.43
7–10-year ETF	0.45	−0.20	−0.06
10–20-year ETF	0.43	−0.33	0.21
>20-year ETF	0.41	−0.40	0.39
B. Performance Measures			
	Mean	Vol	SR
0–1-year ETF	0.26	0.24	1.08
1–3-year ETF	0.79	1.33	0.59
3–7-year ETF	2.07	3.77	0.55
7–10-year ETF	2.87	6.57	0.44
10–20-year ETF	3.01	9.58	0.31
>20-year ETF	4.07	14.02	0.29

Why ETFs? The paper uses UST ETFs as a robustness check to show that the factor structure is not an artifact of implied zeros.

Key interpretation.

- Factor 1 and Factor 2 extracted from ETF excess returns look similar to the implied-zero factor structure.
- This strengthens the interpretation that the first two factors are genuine priced sources of risk in tradable government bond portfolios, not merely artifacts of curve fitting.

7.3 Table 3: GMM estimates of US factor dynamics (baseline, no ZLB shift)

What is being estimated? For each factor $j \in \{1, 2\}$, Table 3 estimates:

- a volatility equation $\sigma_{j,t} = X_t \beta_j^\sigma$ and
- a Sharpe ratio equation $\theta_{j,t} = X_t \beta_j^\theta$,

Table 3: GMM estimates of UST factor dynamics.

	UST Factor 1			UST Factor 2	
	(1a)	(1b)	(1c)	(2a)	(2b)
Volatility Coefficients (β_j^σ)					
Constant	0.51 (1.13)	0.26 (0.57)	0.74 (1.99)	0.13 (1.00)	0.04 (0.32)
Level	2.11 (3.40)	2.50 (3.80)	2.21 (4.18)	0.41 (2.47)	0.42 (2.46)
Slope	4.99 (2.21)	5.66 (2.28)	4.36 (2.84)	2.10 (2.51)	1.89 (2.38)
Curvature	-3.99 (-0.50)	-4.55 (-0.49)		-6.72 (-1.89)	-6.50 (-2.05)
VIX/100	5.13 (3.44)	5.18 (2.88)	4.27 (3.14)	1.47 (2.83)	1.81 (3.60)
Realized Vol	0.44 (0.35)	0.55 (0.37)		2.89 (1.90)	3.97 (2.51)
Sharpe Ratio Coefficients (β_j^θ)					
Constant	-3.07 (-1.19)	-0.46 (-2.04)	-0.42 (-2.11)	2.13 (1.11)	-0.39 (-1.86)
Level	1.91 (1.74)	0.73 (2.40)	0.71 (2.98)	-0.10 (-0.12)	0.83 (2.62)
Slope	4.62 (1.60)	1.87 (1.51)	1.91 (2.61)	-3.85 (-1.09)	0.70 (0.53)
Curvature	-1.76 (-0.27)	0.59 (0.13)		12.92 (1.08)	-1.63 (-0.31)
VIX/100	3.59 (1.47)	1.26 (1.69)	0.71 (1.03)	-0.77 (-0.46)	1.39 (2.19)
Realized Vol	-0.39 (-0.51)	-0.71 (-1.06)		-3.21 (-0.71)	1.18 (0.61)
Return Constant (α_j)	3.87 (0.95)			-1.01 (-1.22)	
No. Moment Conditions	27	27	14	27	27
J-stat p -value (in %)	54.71	45.61	81.47	34.30	12.87
Wald test (1) p -value (in %)	0.01	0.00	0.00	0.04	0.00
Wald test (2) p -value (in %)	43.21	2.73	0.99	41.58	0.97
Goodness-of-fit (1) (in %)	5.45	6.32	6.53	8.85	11.64
Goodness-of-fit (2) (in %)	7.09	6.41	5.03	5.57	1.87

and imposes the return restriction $\mathbb{E}_t[R_{j,t+1}] = \sigma_{j,t}\theta_{j,t}$ via the GMM system.

How I interpret the columns/specifications.

- Multiple specifications correspond to different predictor sets in X_t (e.g., yield-curve variables, VIX, realized-vol persistence).
- The reported “Wald test” p -values test whether volatility is constant (all slope coefficients in the volatility equation are zero) and whether the price of risk is constant (all slope coefficients in the Sharpe equation are zero).

Main takeaways from Table 3.

- **No-arbitrage consistency check:** the factor alphas α_j are economically small and statistically insignificant, supporting the idea that factor risk premia are compensation for priced risk rather than unexplained mean return.
- **Strong evidence of time-varying volatility and time-varying price of risk.** The Wald tests reject constancy in both $\sigma_{j,t}$ and $\theta_{j,t}$ in the US data.
- **Economic reading:** yield-curve state variables and volatility indices help forecast both the quantity of risk and the compensation per unit risk.
- **Practical reading:** the joint model improves GoF in volatility and provides a disciplined decomposition for risk premia, rather than relying on a single “term premium” series.

7.4 Table 4: US factor dynamics allowing ZLB effects

What changes relative to Table 3? Table 4 allows regime dependence via L_t as in (21)–(22):

- γ_j^σ and γ_j^θ capture how predictors map to volatility and Sharpe ratios differently at the ZLB.
- δ_j allows a mean shift in returns during ZLB periods.

How I interpret the results.

- The ZLB can materially alter the *relationship* between volatility and risk premia, i.e., even if volatility rises, the Sharpe ratio need not rise in the same way.
- This matters for policy-era comparisons: the same volatility level may imply different risk compensation depending on monetary-policy constraints.

Table 4: GMM estimates of UST factor dynamics with ZLB effects.

	UST Factor 1		UST Factor 2	
	NonZLB	ZLB Diff	NonZLB	ZLB Diff
Volatility Coefficients (β_j^σ and γ_j^σ)				
Constant	1.25 (2.74)	-1.28 (-1.69)	-0.05 (-0.29)	0.29 (0.98)
Level	0.79 (1.12)	7.85 (1.17)	0.41 (1.63)	-0.91 (-0.39)
Slope	4.76 (2.64)	0.35 (0.11)	3.33 (1.85)	-1.45 (-0.71)
Curvature			-14.09 (-1.89)	8.79 (1.01)
VIX/100	5.14 (2.65)	-1.85 (-0.71)	2.63 (3.41)	-1.90 (-1.63)
Realized Vol			3.07 (1.61)	2.81 (0.70)
Sharpe Ratio Coefficients (β_j^θ and γ_j^θ)				
Constant	-0.47 (-2.11)	-0.29 (-0.62)	-0.38 (-1.60)	-0.10 (-0.21)
Level	0.55 (1.68)	-3.63 (-0.96)	0.87 (2.08)	-2.88 (-0.53)
Slope	1.64 (2.00)	3.77 (1.88)	0.17 (0.08)	1.31 (0.44)
Curvature			2.92 (0.32)	-7.13 (-0.60)
VIX/100	1.53 (1.71)	-1.61 (-1.13)	0.88 (1.13)	1.68 (1.13)
Realized Vol			1.80 (0.77)	-5.97 (-1.07)
No. Moment Conditions	22		29	
J -stat p -value (in %)	47.81		12.01	
Wald test (1) p -value (in %)	0.07		0.06	
Wald test (2) p -value (in %)	1.86		4.10	
Wald test (3) p -value (in %)	57.30		46.17	
Wald test (4) p -value (in %)	12.58		59.18	
Goodness-of-fit (1) (in %)	9.60		13.72	
Goodness-of-fit (2) (in %)	7.29		2.19	

Table 5: R^2 's in monthly and annual return regressions.

A. First-stage volatility regression				
Constant	Level	Slope	VIX/100	R ²
-0.19 (-0.40)	3.08 (4.80)	7.72 (4.78)	6.19 (3.79)	11.82
B. Autoregression of fitted volatility				
Constant	Volhat	R ²		
0.18 (3.03)	0.94 (49.29)	87.19		
C. Second-stage monthly return regression				
Constant	Volhat	R ²		
-1.55 (-2.71) [-3.00]	0.74 (4.02) [4.25]	4.32		
D. Second-stage annual return regression				
Constant	Volhat	R ²		
-10.28 (-4.84) [-2.23]	6.13 (9.06) [4.22]	19.12		
E. BRW-implied annual return regression				
	Volhat	R ²		
	6.43	27.02		

7.5 Table 5: Monthly versus annual R^2 in bond return regressions

What the table is diagnosing. The paper shows that regressions using *overlapping annual returns* often generate mechanically high R^2 when predictors are persistent.

How I interpret the five panels.

- Panels A–C: maturity-specific returns (UST implied zeros, UST ETFs, CGB implied zeros).
- Panels D–E: factor portfolio returns (UST factors, CGB factors).

Key takeaway.

- **Overlapping annual returns inflate R^2 mechanically.** This can mislead interpretation of “predictability strength.”
- **Monthly non-overlapping regressions are cleaner for inference.** They increase sample size and reduce small-sample distortions, especially when predictors (like fitted volatility) are highly persistent.

7.6 Table 6: GMM estimates of China factor dynamics

What Table 6 does. It repeats the Table 3 logic for the Chinese government bond market, estimating $\sigma_{j,t}$ and $\theta_{j,t}$ dynamics for CGB Factors 1 and 2.

Table 6: GMM estimates of CGB factor dynamics.

	CGB Factor 1			CGB Factor 2		
	(1a)	(1b)	(1c)	(2a)	(2b)	(2c)
Volatility Coefficients (β_j^σ)						
Constant	2.24 (2.15)	2.79 (2.68)	3.34 (4.20)	-0.14 (-0.35)	-0.04 (-0.09)	-0.31 (-0.84)
Level	-0.25 (-0.08)	-0.99 (-0.35)	-2.29 (-0.99)	2.06 (1.88)	1.95 (1.70)	2.57 (2.28)
Slope	7.22 (1.71)	4.31 (0.86)		5.66 (4.05)	5.34 (3.92)	5.28 (3.91)
Curvature	-44.88 (-2.85)	-50.98 (-3.18)	-46.48 (-3.07)	-7.02 (-1.04)	-7.35 (-1.01)	
Realized Vol	3.42 (1.90)	2.95 (1.52)	4.06 (2.29)	6.42 (3.18)	4.80 (2.13)	5.14 (2.61)
Sharpe Ratio Coefficients (β_j^θ)						
Constant	-2.89 (-1.81)	-1.34 (-2.32)	-1.15 (-2.67)	1.26 (1.05)	0.83 (1.68)	0.45 (0.95)
Level	3.87 (1.94)	3.88 (2.46)	3.53 (2.72)	-1.90 (-1.12)	-1.19 (-0.84)	-0.27 (-0.20)
Slope	3.49 (0.92)	0.73 (0.32)		-4.80 (-1.50)	-3.93 (-2.15)	-4.25 (-2.32)
Curvature	-5.09 (-0.22)	21.28 (2.00)	21.10 (1.95)	-13.61 (-1.74)	-11.15 (-1.53)	
Realized Vol	2.81 (2.04)	1.24 (1.26)	1.22 (1.35)	1.47 (0.49)	2.30 (1.25)	1.26 (0.83)
Return Constant (α_j)	3.55 (1.18)			-0.12 (-0.22)		
No. Moment Conditions	20	20	14	20	20	14
J-stat p -value (in %)	14.34	24.72	8.99	57.51	59.58	66.19
Wald test (1) p -value (in %)	1.99	0.96	0.31	0.00	0.00	0.00
Wald test (2) p -value (in %)	3.50	1.43	1.39	0.49	1.99	4.67
Goodness-of-fit (1) (in %)	8.68	5.79	4.91	16.20	15.26	14.08
Goodness-of-fit (2) (in %)	7.05	6.88	6.32	6.14	6.46	4.26

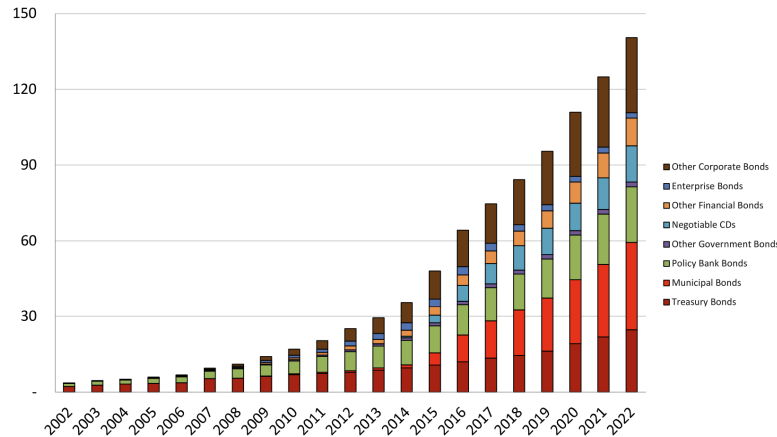


Fig. 1. China bond market capitalization in trillion CNY.

What I learn from Table 6.

- **Time variation exists in China as well:** both volatility and price-of-risk dynamics are statistically important.
- **But the co-movement differs:** the paper emphasizes that the correlation between the fitted volatility and fitted Sharpe ratio is *negative* for China factors (in contrast to the US).
- **Interpretation:** in China, episodes of higher “risk” (volatility) do not necessarily coincide with higher compensation per unit risk, consistent with institutional features (policy interventions, segmentation, or non-market pricing components).

8 Interpreting the figures (Fig. 1–Fig. 8)

8.1 Fig. 1: China bond market capitalization

What I see. A long-run increase in China’s bond market size (in CNY), with composition across categories.

Why it matters for the paper.

- It motivates studying China as a major, rapidly growing government bond market.
- Growth and structural change can affect liquidity, participation, and how risk is priced, which is consistent with the later evidence that price vs. quantity of risk behave differently than in the US.

8.2 Fig. 2: MOVE index (US interest rate volatility)

What I see. Large time variation with spikes around crisis episodes.

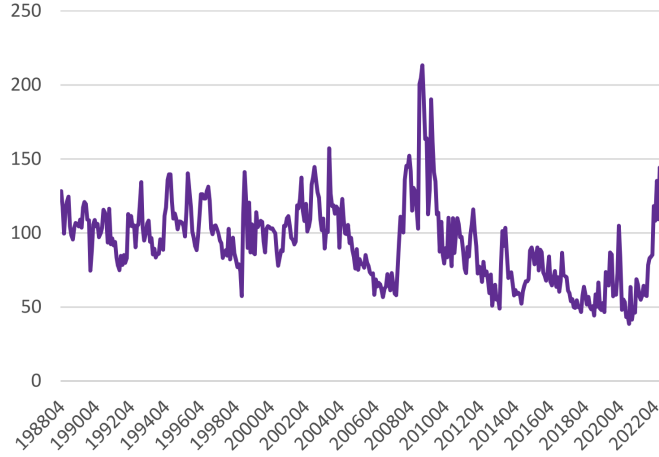


Fig. 2. MOVE index of implied yield volatility from one-month Treasury bond options.

Why it matters.

- It provides immediate external evidence that bond-market volatility is far from constant.
- This supports the modeling choice where $\sigma_{j,t}$ is time-varying and forecastable.

8.3 Fig. 3: US Factor 1 dynamics (volatility, Sharpe ratio, risk premium)

How to read it.

- Top: fitted $\sigma_{1,t}$ (quantity of risk).
- Middle: fitted $\theta_{1,t}$ (price of risk / Sharpe ratio).
- Bottom: fitted risk premium $\sigma_{1,t}\theta_{1,t}$.

My interpretation.

- Factor 1 volatility and Sharpe ratio tend to move together in the US (positive correlation), generating large swings in risk premia.
- This is a concrete visualization of the paper’s main decomposition: the risk premium is not only “volatility” and not only “expected return”; it is the *interaction* of the two.

8.4 Fig. 4: US Factor 2 dynamics

Interpretation relative to Factor 1.

- Factor 2 dynamics illustrate that different priced sources of risk can have different volatility and Sharpe ratio behavior.

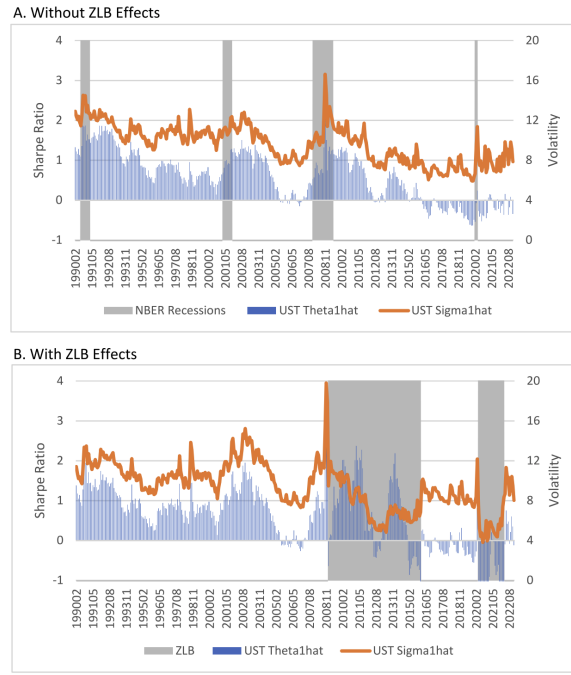


Fig. 3. UST Factor 1 dynamics.

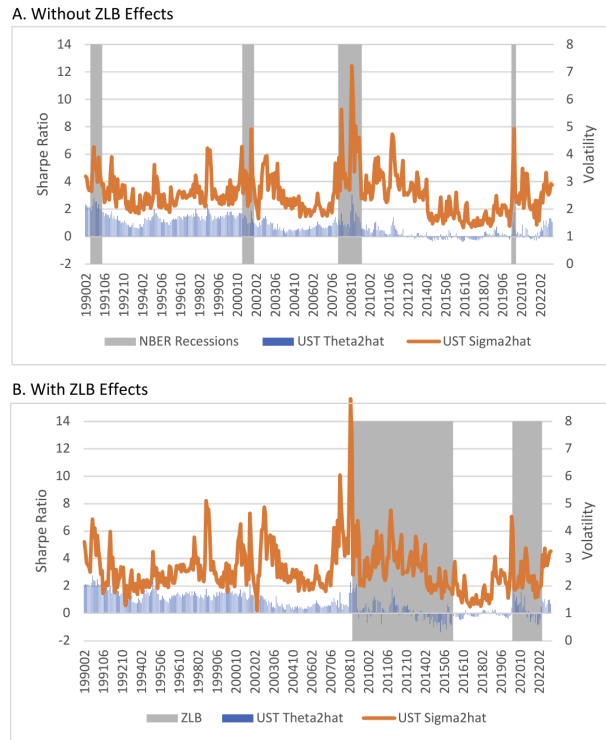


Fig. 4. UST Factor 2 dynamics.

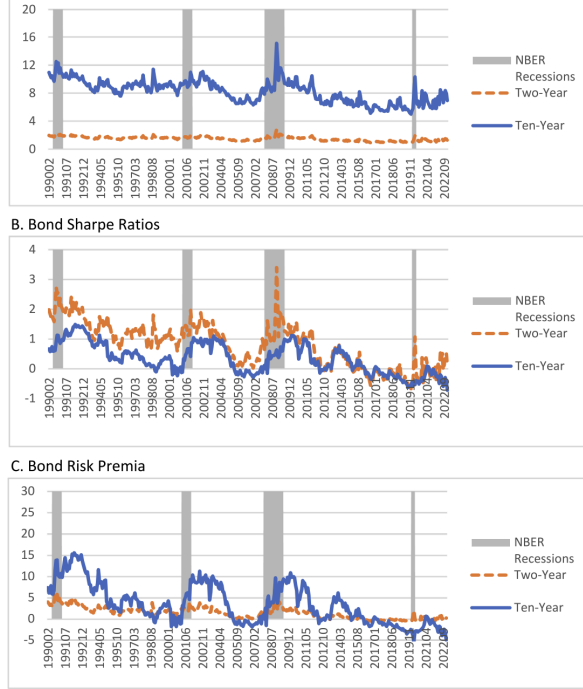


Fig. 5. UST bond dynamics.

- Since maturity portfolios load differently on Factor 2 (positive for short, negative for long), Factor 2 dynamics help explain maturity patterns in Sharpe ratios and risk premia.

8.5 Fig. 5: US bond dynamics (2-year and 10-year implied zeros, baseline)

What is plotted. For each of 2-year and 10-year implied zeros:

- annualized fitted volatility,
- fitted Sharpe ratio,
- fitted risk premium.

These objects are computed from factor estimates using (31)–(33).

Key insight.

- The maturity-specific risk premium inherits the volatility–Sharpe interactions of the two factors.
- Therefore, a “term premium” interpretation that ignores whether changes come from σ or θ can be incomplete.

8.6 Fig. 6: US bond dynamics with ZLB effects

What changes.

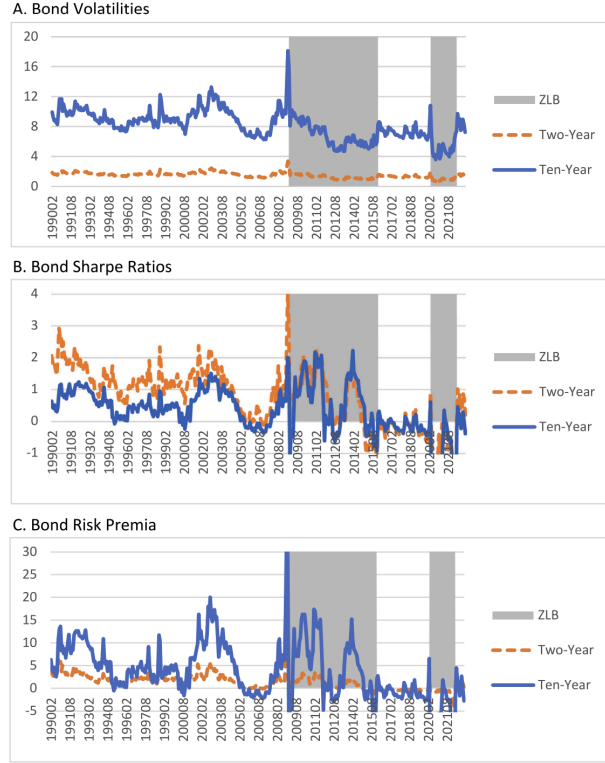


Fig. 6. UST bond dynamics with ZLB effects.

- Allowing ZLB regime dependence can change the fitted Sharpe ratio dynamics, especially during periods when conventional monetary policy is constrained.
- This visually reinforces why Table 4 matters: the mapping from state variables to expected returns is not stable across regimes.

8.7 Fig. 7: China Factor 1 and Factor 2 dynamics

Main contrast with US.

- The paper emphasizes **negative correlations** between fitted volatility and fitted Sharpe ratios in China factor portfolios.
- This implies that increases in volatility may coincide with decreases in Sharpe ratios (and vice versa), so risk premia do not mechanically track volatility.

8.8 Fig. 8: China bond dynamics (2-year and 10-year implied zeros)

What it shows.

- Maturity-specific volatility, Sharpe ratio, and risk premia time series for Chinese bonds.

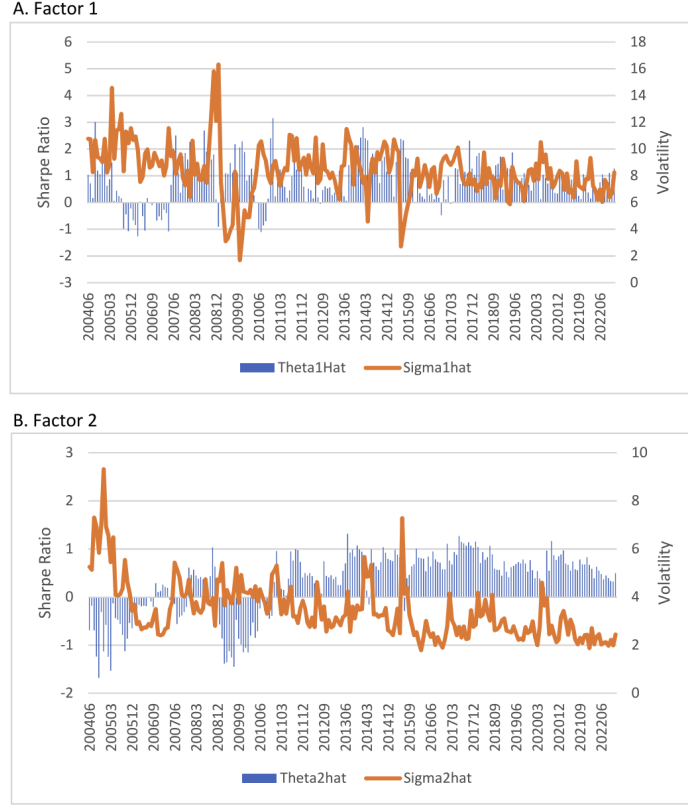


Fig. 7. CGB factor dynamics.

- The negative price–quantity co-movement at the factor level can feed into maturity-specific dynamics in a way that differs from the US.

9 Concluding synthesis (my coherent mental model)

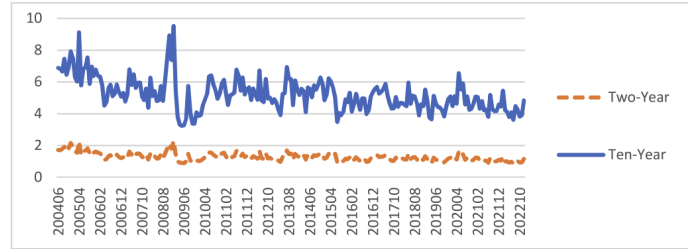
9.1 What I would say if asked “what is the paper really doing?”

- It builds a disciplined **risk-premium decomposition** for government bonds:

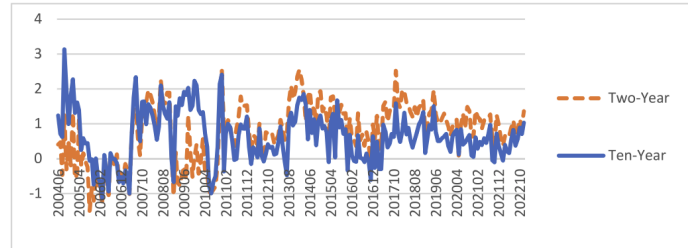
$$\text{risk premium} = \underbrace{\text{volatility}}_{\text{quantity of risk}} \times \underbrace{\text{Sharpe ratio}}_{\text{price of risk}}.$$

- It **estimates** both objects using a joint system (returns + volatility proxy) that encodes the decomposition restriction.
- It **compares** the time-series behavior of these objects across the US and China, showing that the joint dynamics of risk and compensation are structurally different.

A. Bond Volatilities



B. Bond Sharpe Ratios



C. Bond Risk Premia

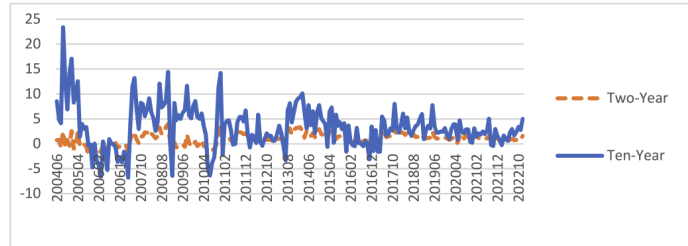


Fig. 8. CGB bond dynamics.

Coherent Summary (What the paper does)

The paper studies government bond *risk and return* in the US Treasury (UST) market and the Chinese government bond (CGB) market using a framework inspired by the empirical *equity* risk–return literature rather than the traditional term-structure literature that emphasizes closed-form bond pricing models.

The empirical object is not “the yield curve” per se, but the joint time-series behavior of:

- the **quantity of interest-rate risk** (conditional volatility), and
- the **price of interest-rate risk** (conditional Sharpe ratio),

for principal-component bond-factor portfolios constructed from excess returns across maturities. The paper estimates these objects in a way that remains consistent with no-arbitrage restrictions but avoids imposing tight functional forms that often come with closed-form term-structure models.

The decomposition that organizes everything

A single identity structures the interpretation throughout:

$$\underbrace{\mathbb{E}_t[R_{t+1}^e]}_{\text{conditional risk premium}} = \underbrace{\sigma_t}_{\text{quantity of risk}} \cdot \underbrace{\theta_t}_{\text{price of risk (Sharpe)}},$$

where the paper estimates σ_t and θ_t as state-dependent objects for factor portfolios, and then maps them back into maturity-specific bond risk premia.

Main conclusions

1. **Two priced bond factors matter in both countries.** When the analysis is organized around Sharpe ratios, government bond returns in both the US and China exhibit two empirically important principal-component factors with priced risk premia.
2. **Both the quantity and the price of risk are strongly time-varying.** For both factors and in both markets, conditional volatility and conditional Sharpe ratios vary substantially over time, implying that variation in risk premia cannot be understood by looking only at volatility or only at expected returns.
3. **Factor structure is similar across US and China despite segmentation, but the dynamics differ.** The factor structure of risk premia in China is broadly similar to the US even though the Chinese bond market was effectively segmented from the US market for much

of the sample, suggesting these factors reflect fundamental components of bond risk premia rather than purely local artifacts. However, the *co-movement* between the price and quantity of risk differs meaningfully across time and across countries.

4. **The correlation between price and quantity of risk is not stable (and can flip sign).** In some periods the price of risk and the quantity of risk move together, but in other periods—notably the US ZLB era and episodes of major policy intervention in China—the correlation becomes much lower or even negative. This instability is central: it implies that a “one-size-fits-all” restriction linking σ_t and θ_t is empirically unreliable.
5. **Implication for theory: closed-form term-structure models can be too restrictive.** Because many closed-form bond-pricing models impose tight restrictions on the relation between the price and quantity of risk, the documented time variation (including sign changes) is difficult to reconcile with those frameworks. A good fit in post-Volcker US data should therefore not be interpreted as evidence that such models are flexible enough to capture default-free bond returns in general.

Intuition (Why the results make sense)

Intuition 1: why “risk premia” must be decomposed

If risk premia satisfy $\mathbb{E}_t[R_{t+1}^e] = \sigma_t \theta_t$, then changes in expected bond returns can arise from:

- changes in uncertainty (*volatility* σ_t), and/or
- changes in compensation per unit risk (*Sharpe ratio* θ_t).

Looking only at a term premium series (or only at volatility) collapses these two distinct economic forces into one number and can therefore hide the mechanism.

Intuition 2: why the correlation between σ_t and θ_t matters

The sign and stability of $\text{Corr}(\sigma_t, \theta_t)$ governs whether volatility is a reliable proxy for expected returns:

- If σ_t and θ_t are **positively** correlated, higher volatility tends to coincide with higher Sharpe ratios, amplifying swings in risk premia.
- If they are **negatively** correlated, volatility spikes can coincide with lower Sharpe ratios, meaning risk premia may not rise much (or may even fall) when risk rises.

Hence, cross-country and regime-specific differences in this correlation are economically first-order.

Intuition 3: policy regimes can “decouple” risk from compensation

The paper highlights that aggressive monetary-policy episodes can coincide with a decoupling of risk quantity and risk pricing. One plausible intuition is:

- policy interventions can compress or stabilize yield dynamics (affecting σ_t),
- while simultaneously changing investors’ required compensation (affecting θ_t),

and these two effects need not move in the same direction.

Additional methodological takeaway: why the paper downplays annual overlapping-return R^2

The paper stresses that using monthly overlapping annual returns can mechanically inflate predictive regression R^2 when predictors are persistent, creating the illusion of strong predictability. Monthly non-overlapping returns are cleaner for inference and avoid these mechanical boosts.

Bottom line

The paper concludes that understanding government bond risk premia requires a flexible no-arbitrage-consistent decomposition into time-varying quantities and prices of risk. While the basic factor structure looks similar in the US and China, the time variation in the relation between volatility and Sharpe ratios—especially in policy-constrained or intervention-heavy periods—poses a challenge for widely used closed-form term-structure models and motivates more flexible empirical and theoretical approaches.